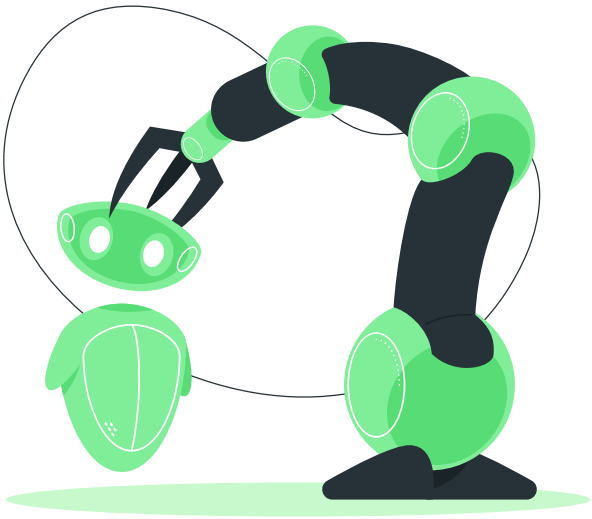
Robotics
CS 3015

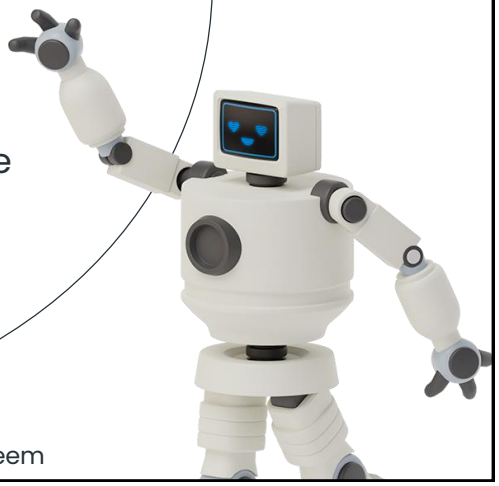
Umarah Qaseem



1

2a

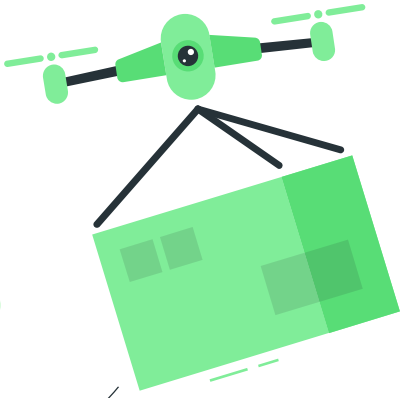
- Position in the world
- Frame of Reference
- Refresher 2D Geometry
- Story of Cartesian Plane
- Points and Vectors
- Convention of coordinate frame
- Pose illustrations
- Pose 2D
- Relative Positions
- Relative Pose



2

Home Task?

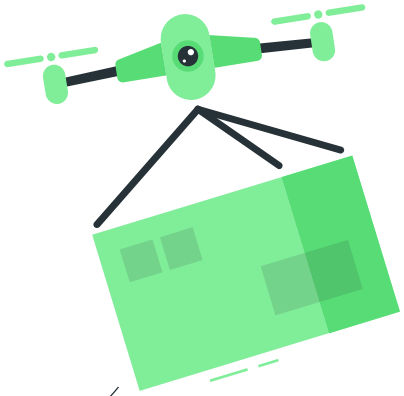
What are three lessons learned from the movie?



3

What will we do today?

Position of things in the world



Revision

4



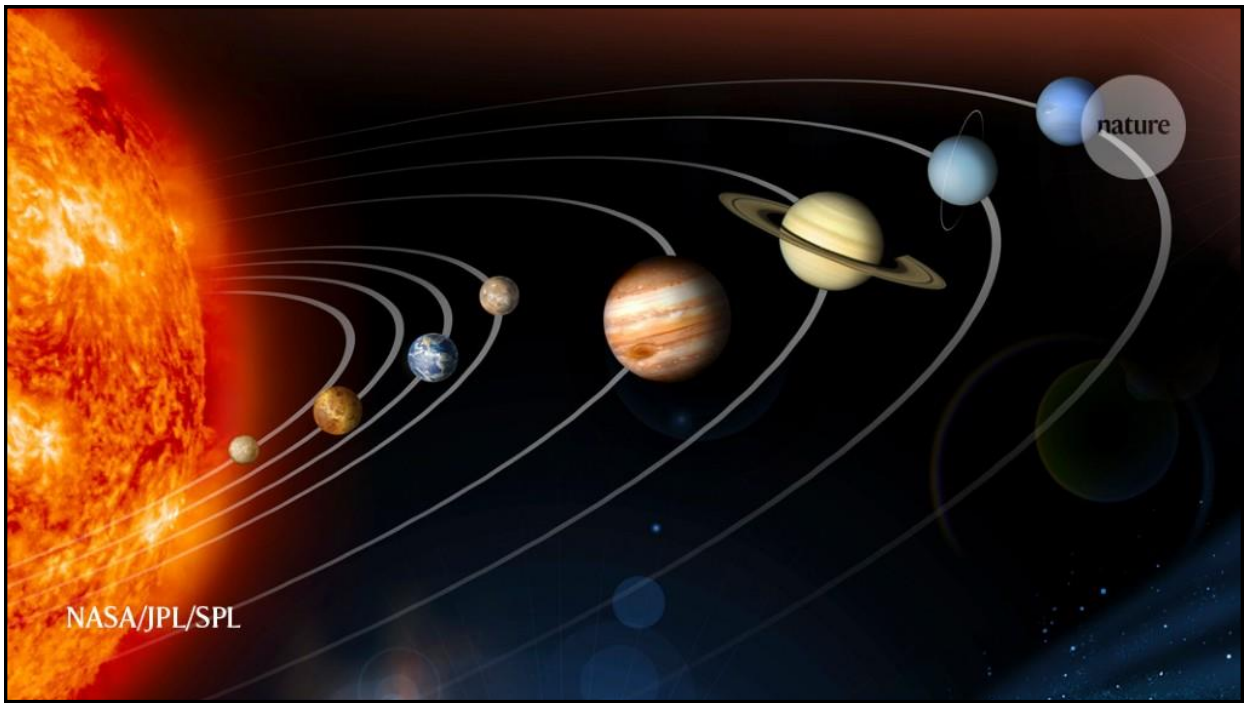
Frame of Reference

For describing position, we need;

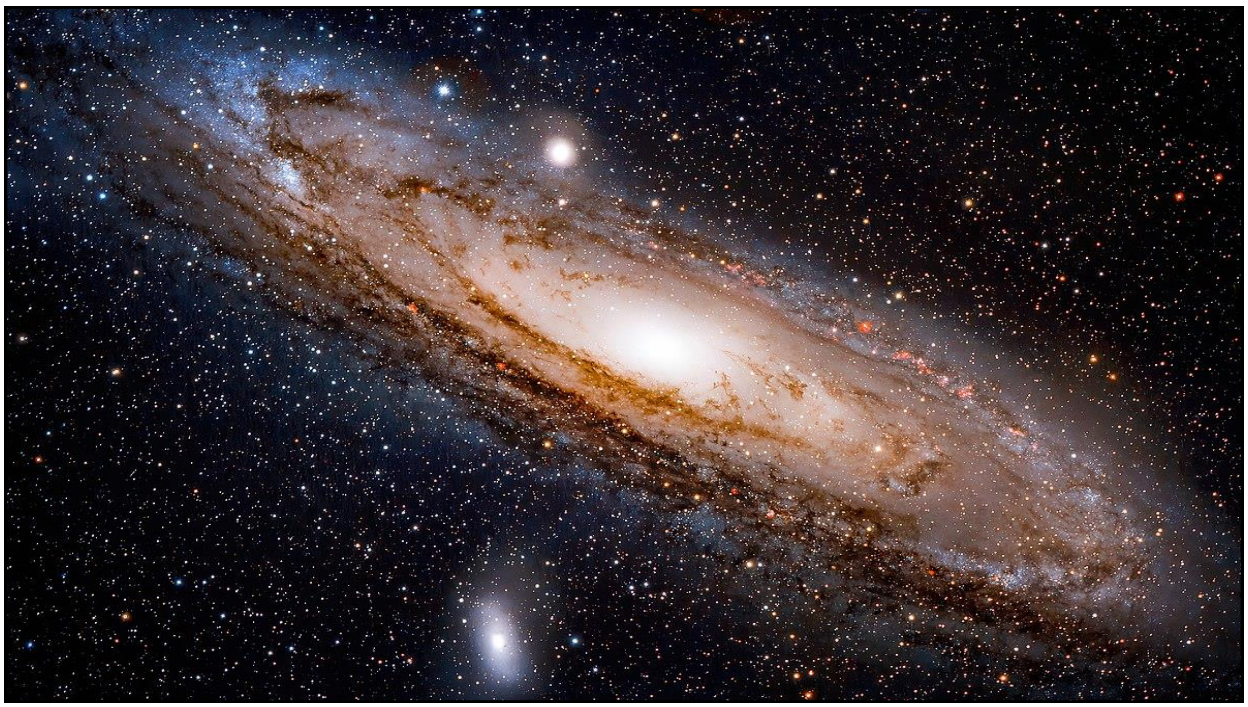
5



6



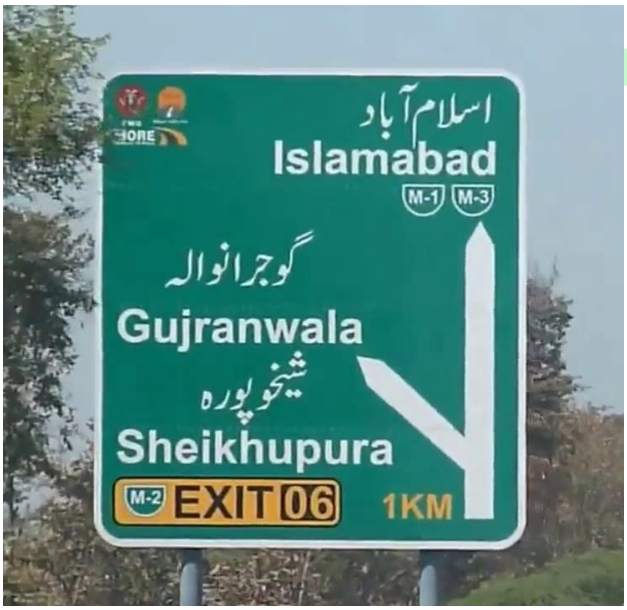
7



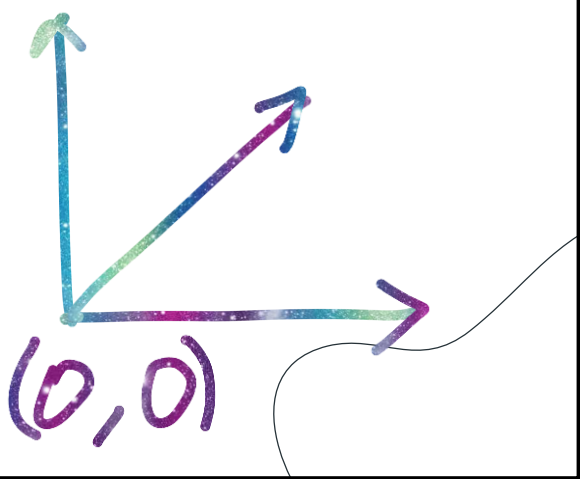
8



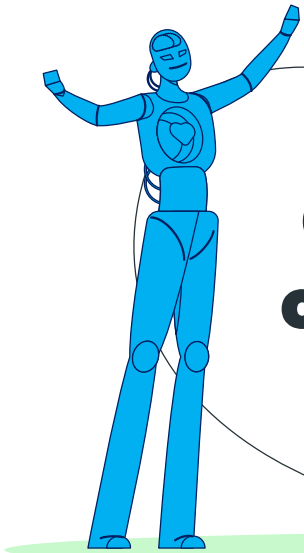
9



So put a stake in ground!



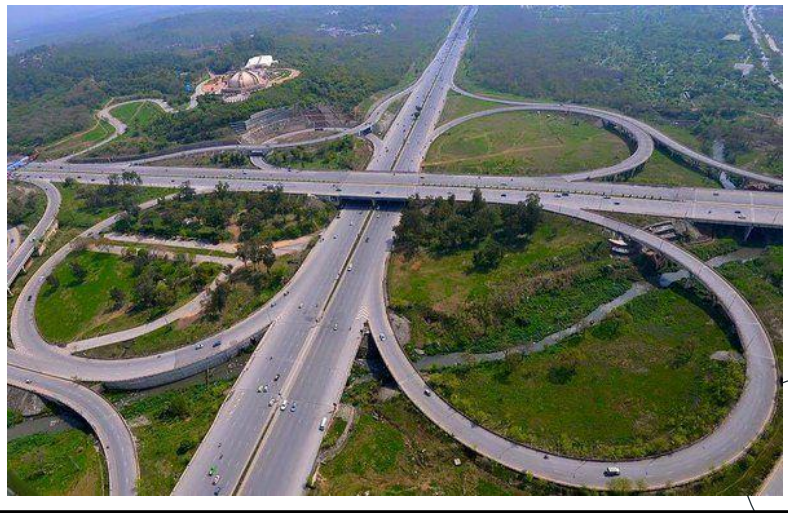
10



Create your own frame of Reference!

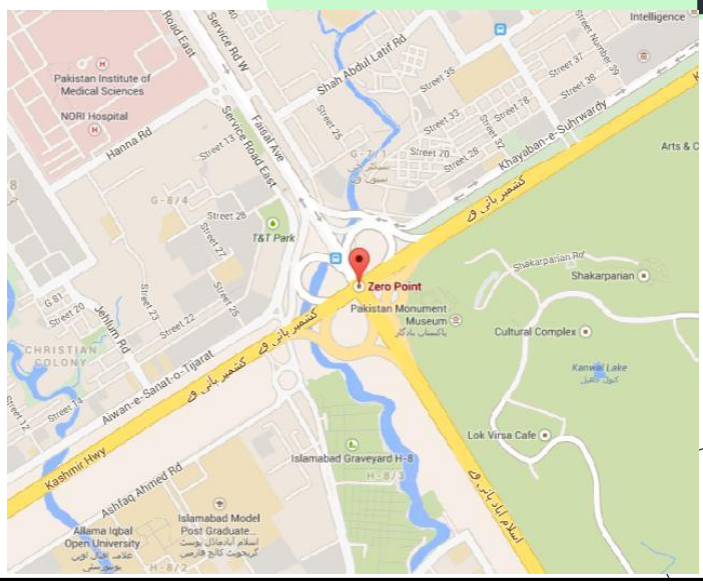
11

Do you know Islamabad's point of Reference?



12

Zero point



13

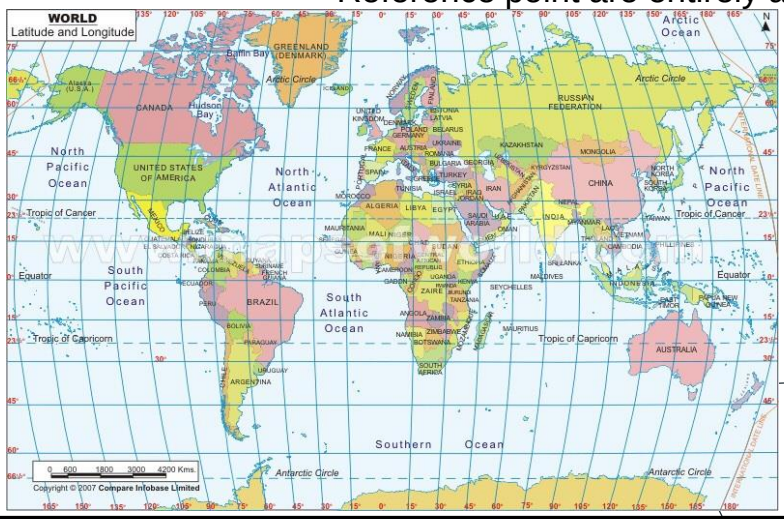
Position in 2D

- X axis
- Y axis

14

Reference System on our Planet

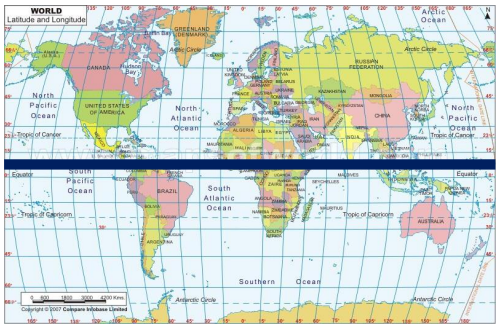
Reference point are entirely arbitrary.



15

Equator

A plane Orthogonal to rotational axis.

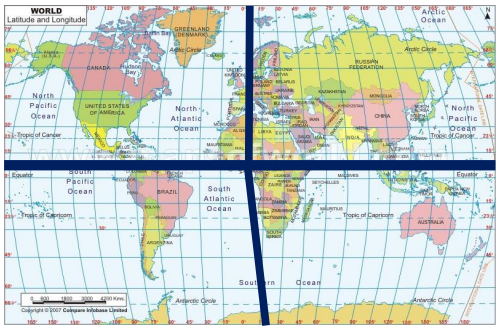




16

Prime Meridian

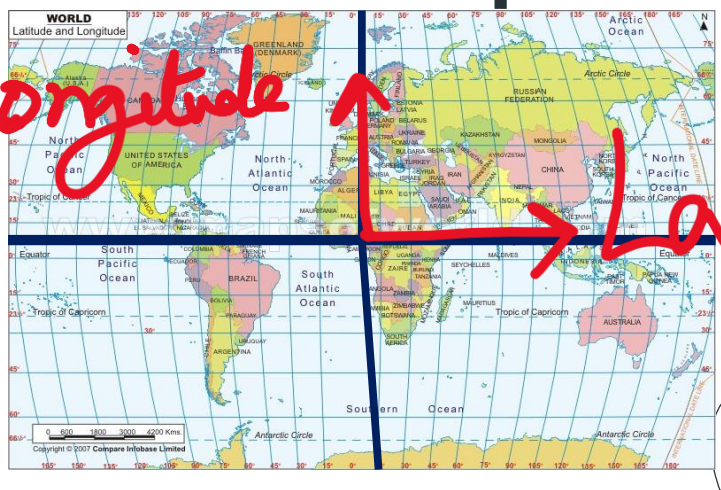
.





17

They are enough to describe any point on earth



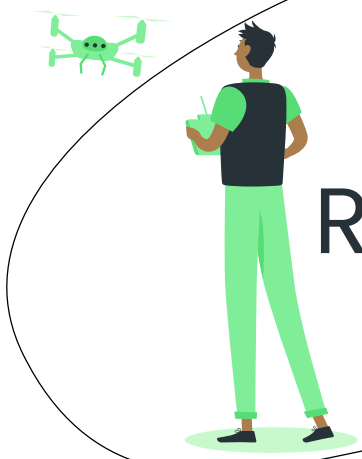
18

Conclusion

These are completely arbitrary lines. We need them to describe locations/points on earth.

Now you know why we need frame of reference!

19



Refresher 2D Geometry

Robotics – Umarah Qaseem

20

Euclidean Geometry

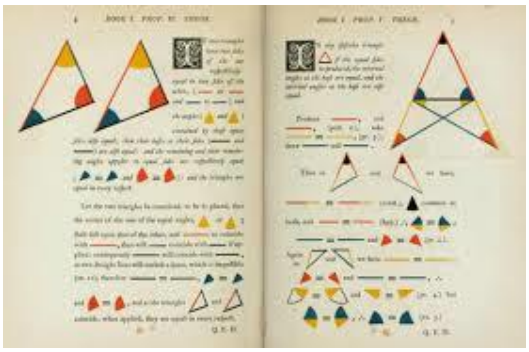


- Euclid of Alexandria 300 BCE
- The Elements (13 books)
- Basics of Geometry



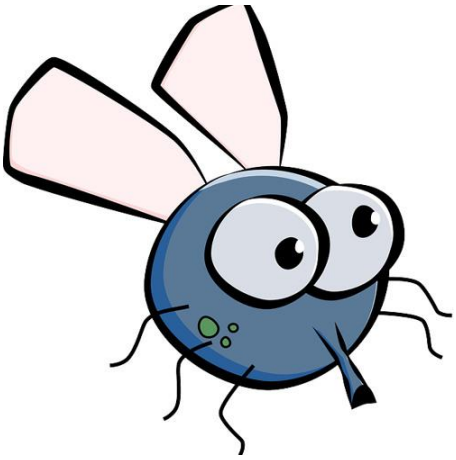
21

Euclidean Geometry



- Geometry of Plane and 3D space
- line, triangles circles
- No numbers

22



Rene Descartes

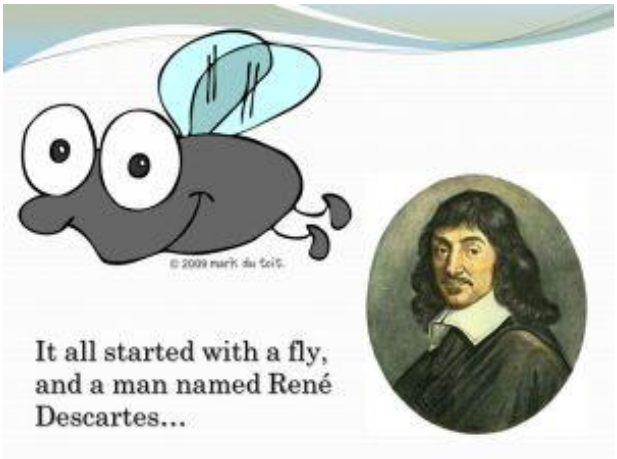
- Philosopher
- Mathematician
- Father of Modern Philosophy
- Cogito ergo sum
- “I think, therefore I am”



23

Story of Cartesian Plane

- Once upon a time…



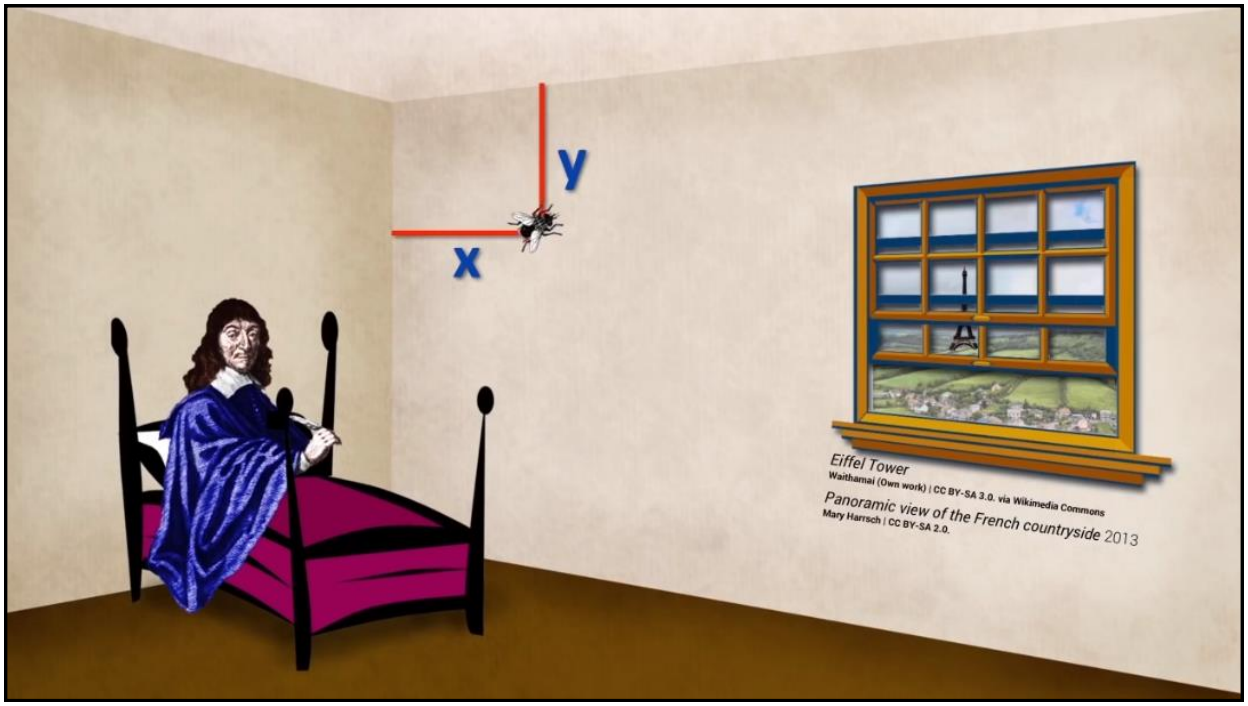
24


Eiffel Tower
Waithamai (Own work) | CC BY-SA 3.0. via Wikimedia Commons
Panoramic view of the French countryside 2013
Mary Harrsch | CC BY-SA 2.0.

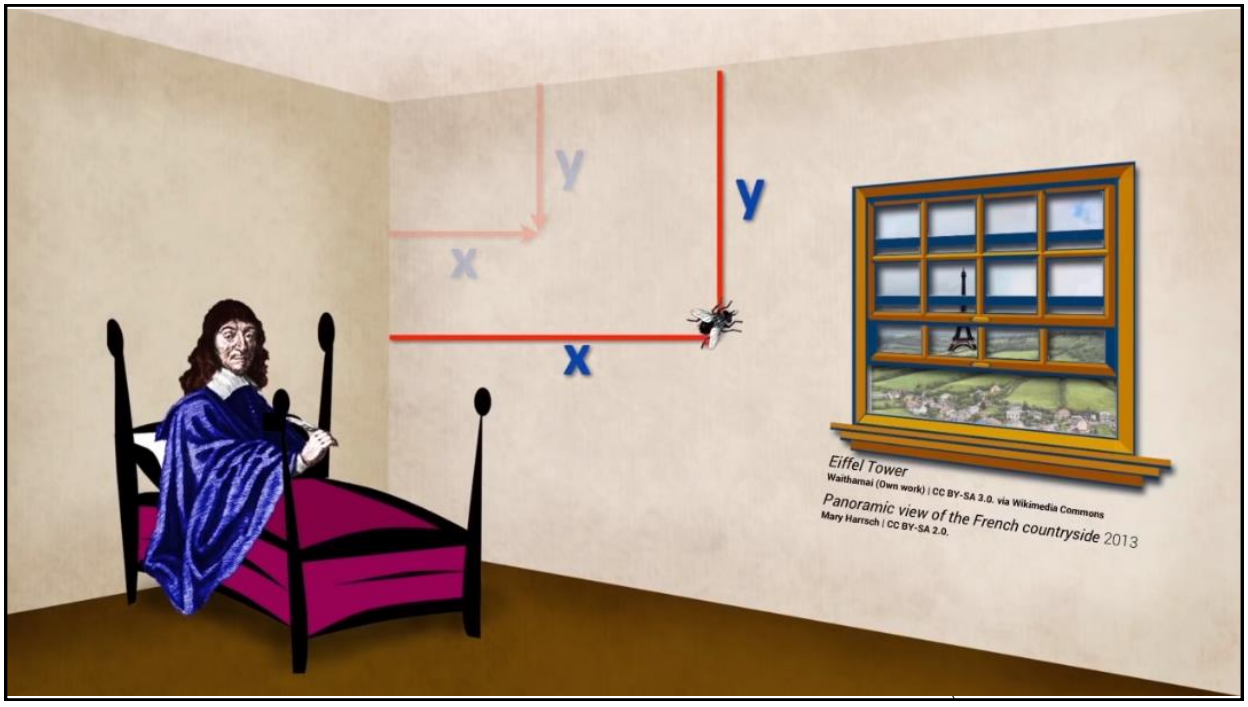
25


Eiffel Tower
Waithamai (Own work) | CC BY-SA 3.0. via Wikimedia Commons
Panoramic view of the French countryside 2013
Mary Harrsch | CC BY-SA 2.0.

26

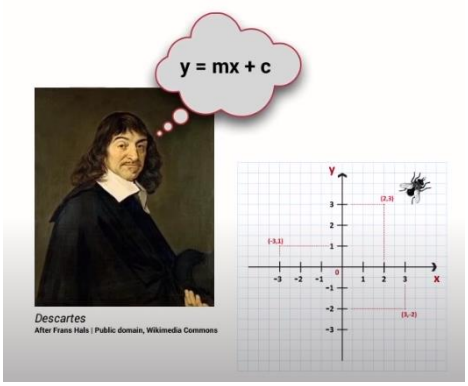

y
x
Eiffel Tower
Waithamai (Own work) | CC BY-SA 3.0. via Wikimedia Commons
Panoramic view of the French countryside 2013
Mary Harrsch | CC BY-SA 2.0.

27


y
x
y
x
Eiffel Tower
Waithamai (Own work) | CC BY-SA 3.0. via Wikimedia Commons
Panoramic view of the French countryside 2013
Mary Harrsch | CC BY-SA 2.0.

28

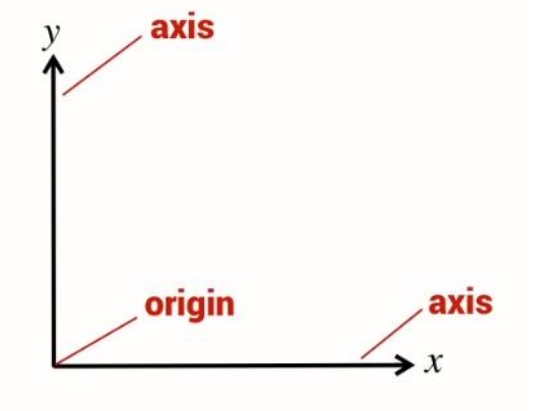
Cartesian Geometry



- Position of something in 2D frame can be described using 2 coordinates
- Added algebra to Euclidean geometry
- Shapes could be expressed by equations
- Cartesian or analytic geometry
- Coordinate system

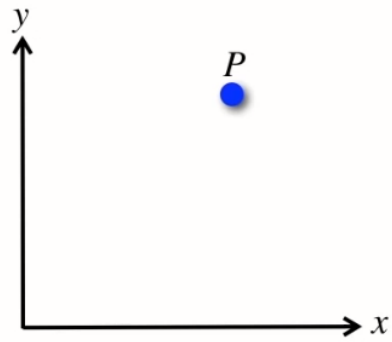
29

2D Coordinate Frame



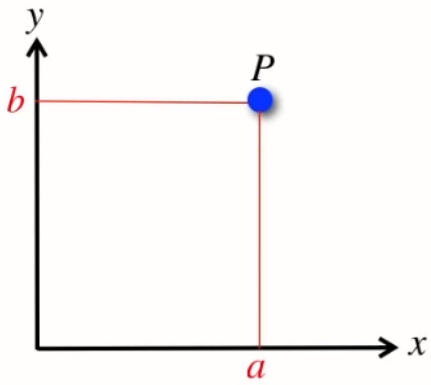
30

Points



31

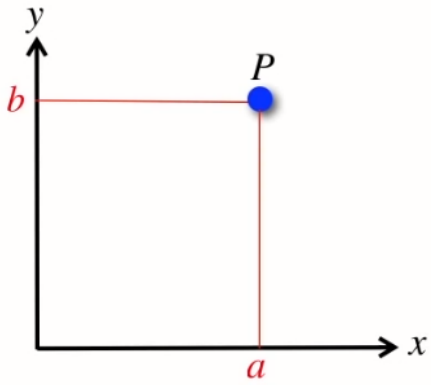
Points



$$P = (a, b)$$

32

Points

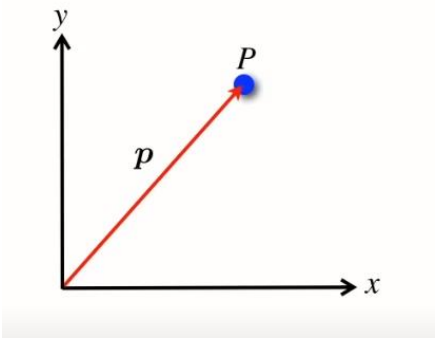


$$P = (a, b)$$

- A point is a location in n dimensional space $P \in \mathbb{R}^n$
- Represented by a cartesian coordinate or n-tuple ($x_1, x_2, \ldots x_n$)

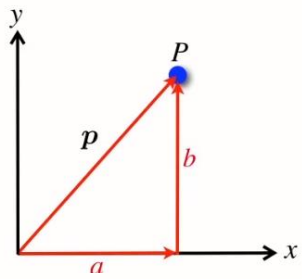
33

Vectors



34

Vectors



$$\boldsymbol{p} = \begin{bmatrix} a \\ b \end{bmatrix}$$

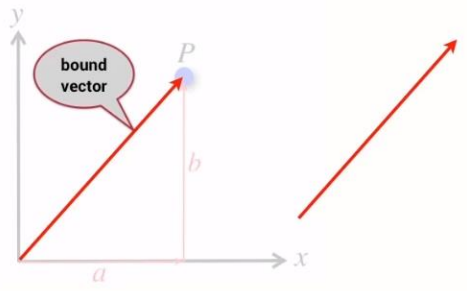
A vector is an element of an n-dimensional vector space $\boldsymbol{p} \in \mathbb{R}^n$

Represented by an n-element column vector $[x_1, x_2 \cdots x_n]^T$

Can be considered a relative displacement

35

Vectors



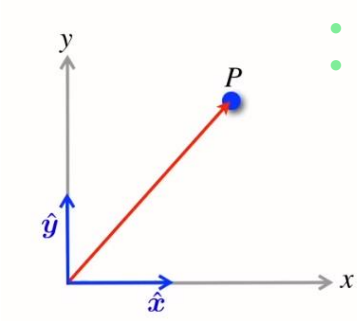
The vector has a particular starting point

36

Unit Vectors

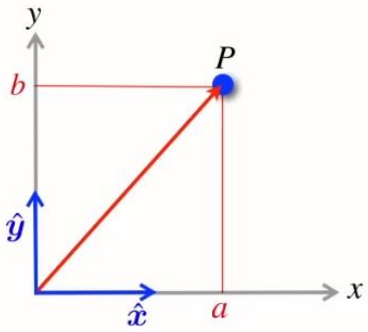
- A vector with unit length
- Length of 1

$$\boldsymbol{p} = \begin{bmatrix} a \\ b \end{bmatrix}$$



37

Unit Vectors



- A vector with unit length
- Length of 1

$$\boldsymbol{p} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\boldsymbol{p} = a\hat{\boldsymbol{x}} + b\hat{\boldsymbol{y}}$$

38

Points and Vectors

- Points:

- Define a location
- Cannot add or multiply points

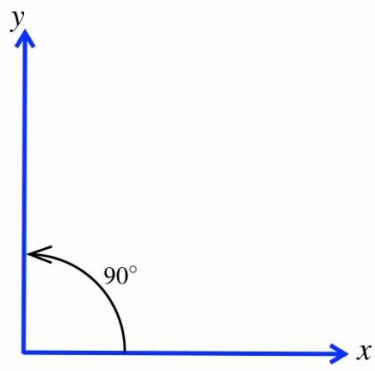
- Vectors:

- Do not define a location
- Rather they specify how to get from one location to another
- Can add and subtract
- Can multiply with a scaler

- The difference between two points is a vector
- A point is defined by a vector displacement from another point (normally the origin of a coordinate frame)
- Both can be represented by n real numbers.

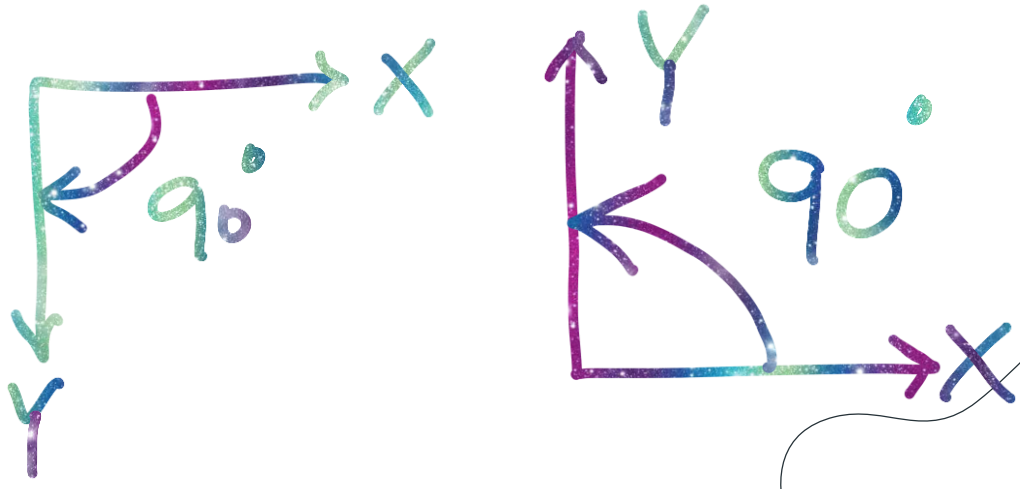
39

Right-handed coordinate frame



40

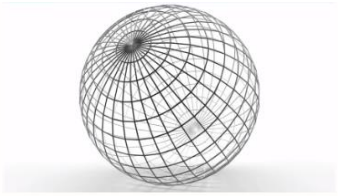
Right-handed coordinate frame



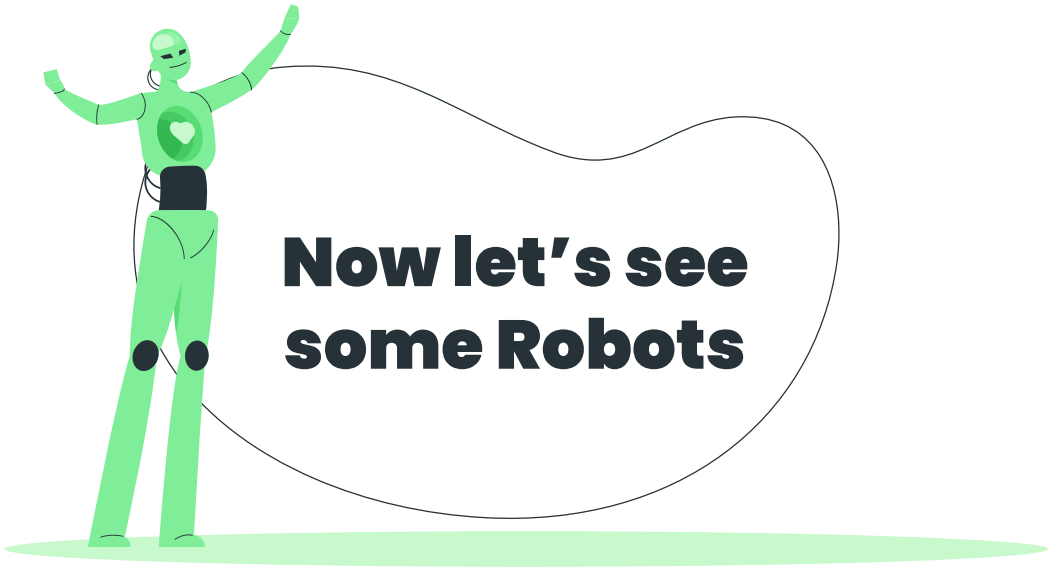
41

Euclid's geometry Applies to planes

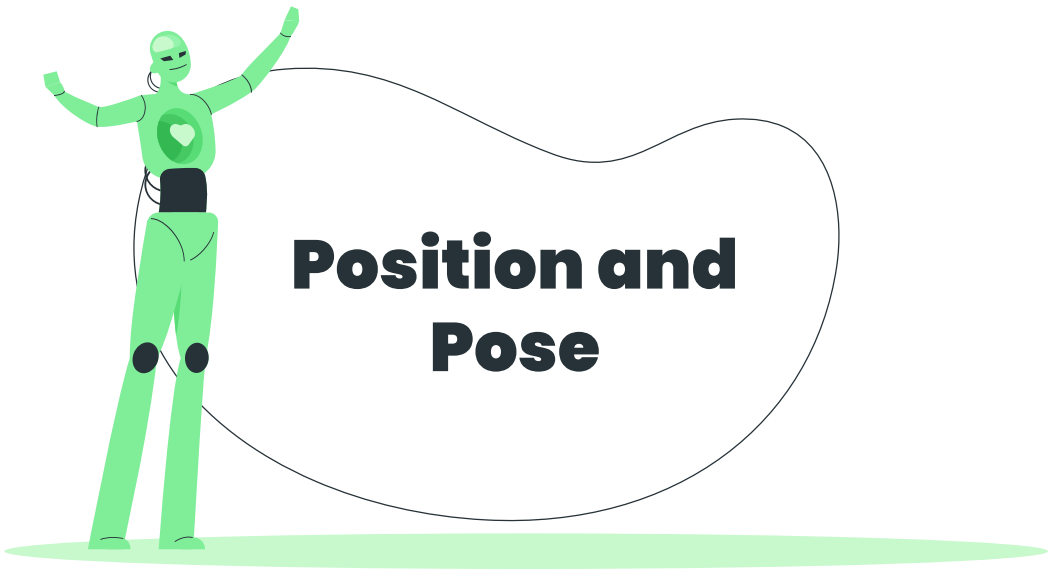
- Parallel lines never intersect (Postulate 5)
- However, on surface of earth, parallel lines at equator will intersect at poles.
- So, it is a non-Euclidean geometry
- Euclidean Geometry is a good approximation over small areas may be in a room or city



42



43



44

Illustrations of POSE through toy robots in class

Robotics – Umarah Qaseem

45



Pose

Pronounced as ksi

It has 3 parameters

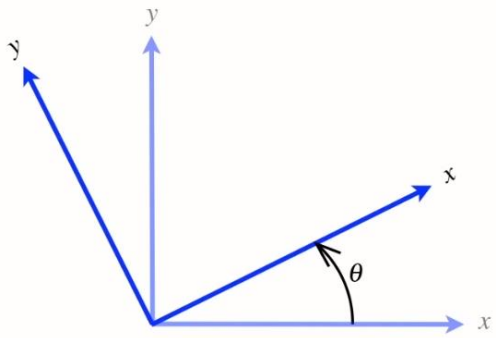
$$(x, y, \theta)$$

$$\xi$$

48

Coordinate Frame Rotation convention

- Angles increase positively in anti clockwise direction



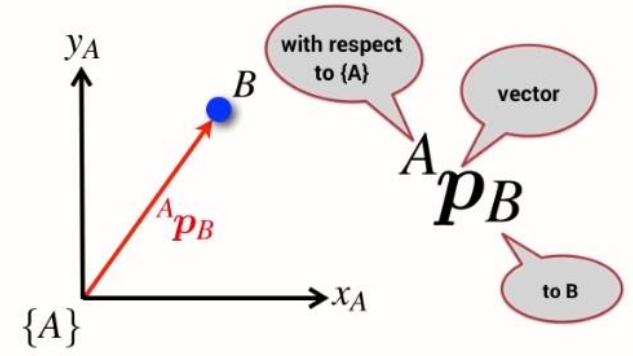
Pose

- Pose has 3 parameters
- Think of it as a move (x, y) and rotation/twist (θ)

$$\xi$$

Point and Vector

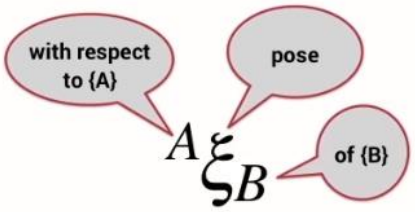
- Point is described by a vector with respect to a coordinate frame.

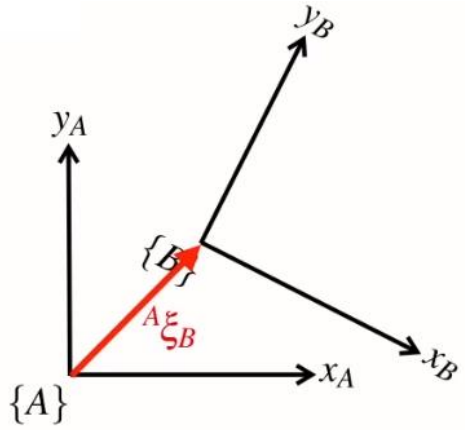


51

Pose

- Pose of a coordinate frame can be described with respect to another coordinate frame.
- Pose can be considered as the motion of a coordinate frame
  - A translation and a rotation





52

Relative Positions



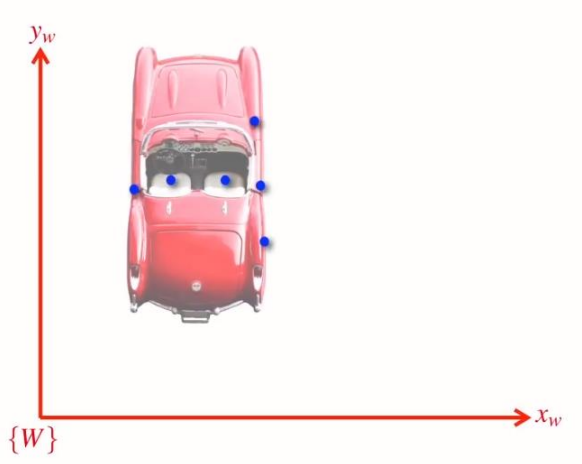
53

A Real World Scenario



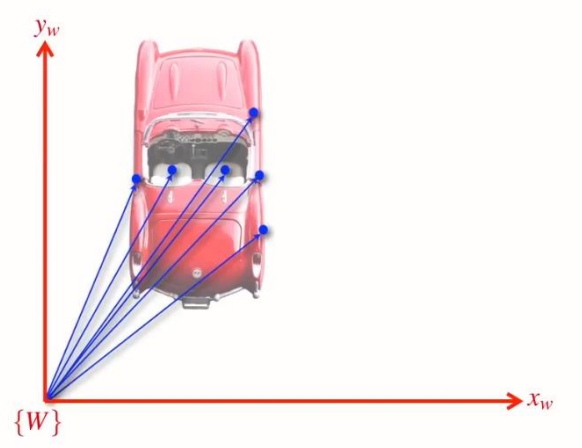
54

A Real World Scenario



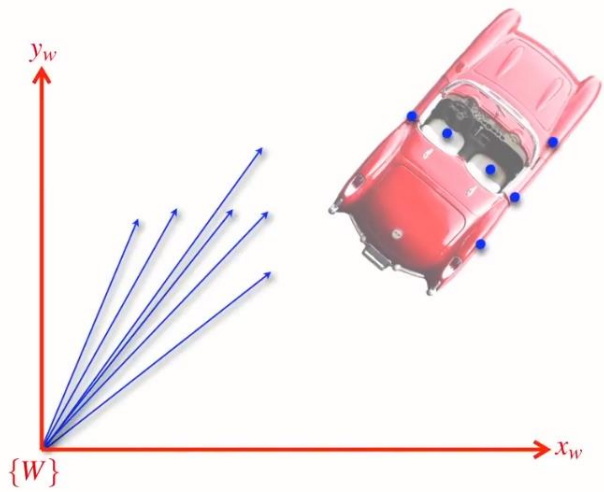
55

A Real World Scanerio



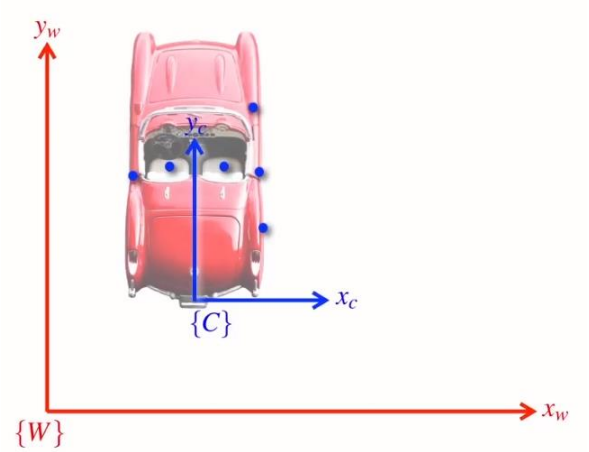
56

A Real World Scenario



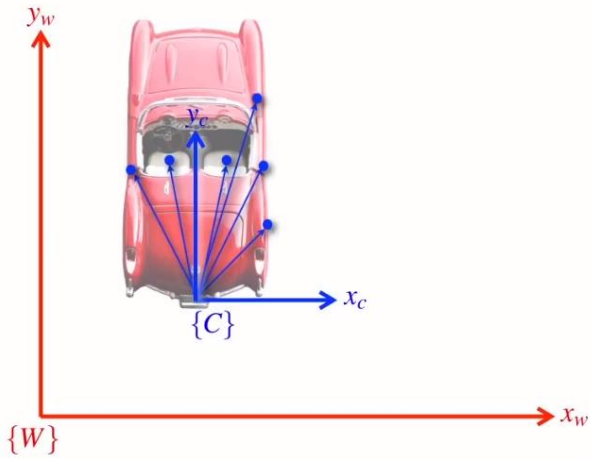
57

A Real World Scenario



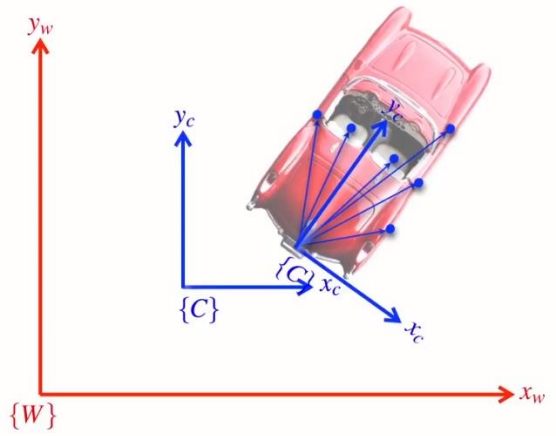
58

A Real World Scenario



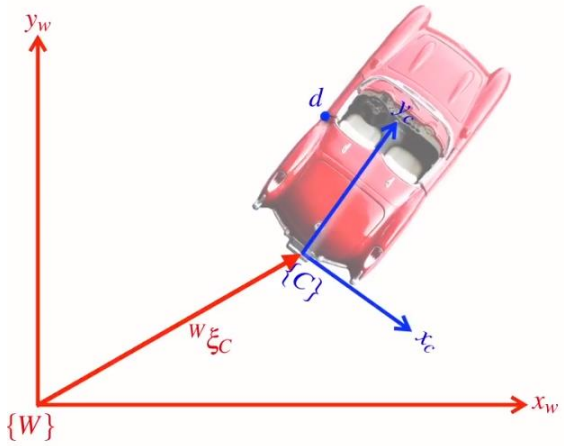
59

A Real World Scenario



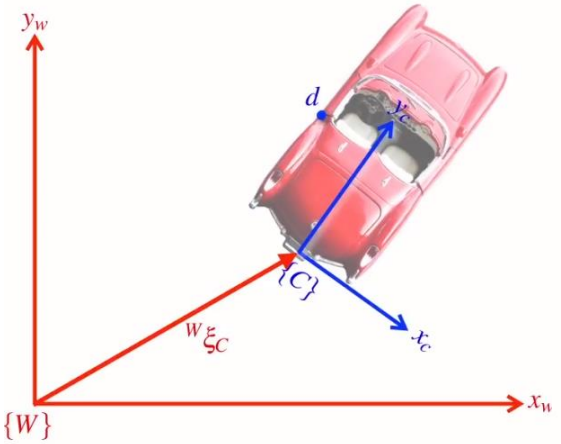
60

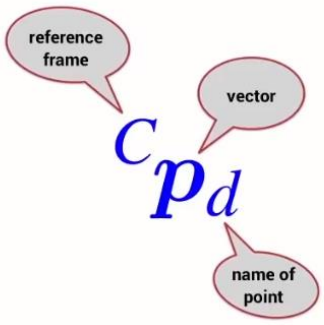
A Real World Scenario



61

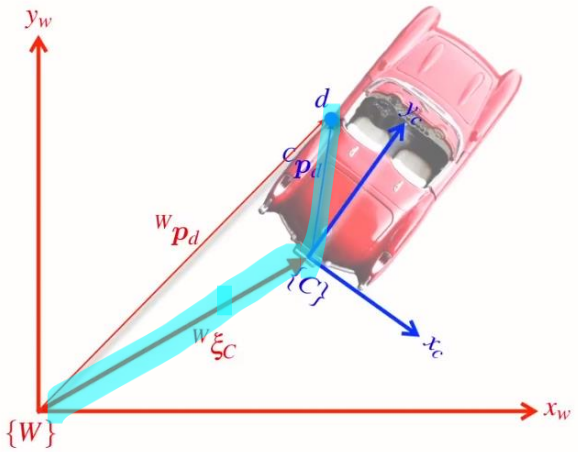
A Real World Scenario





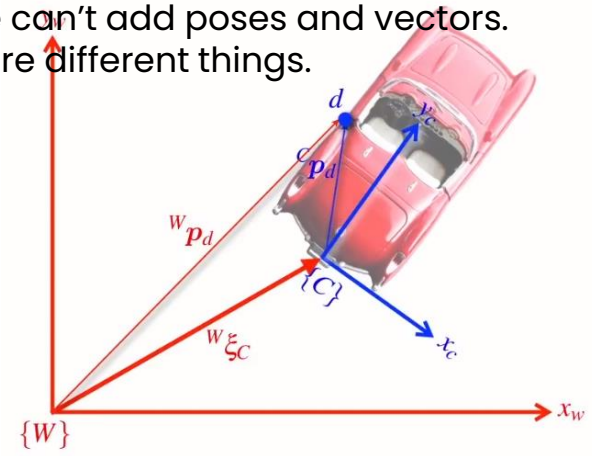
62

A Real World Scenario



A Real World Scenario

- But we can't add poses and vectors.
- They are different things.

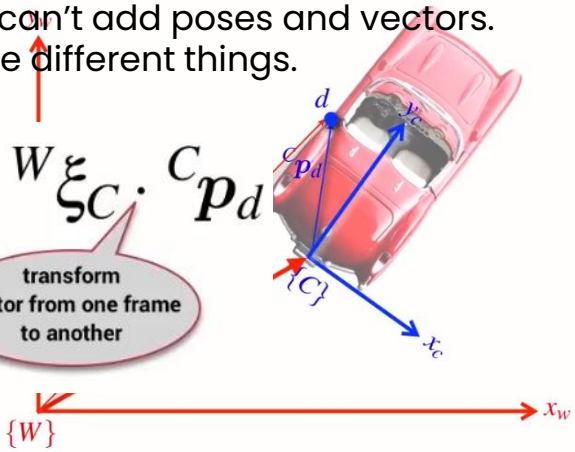


A Real World Scenario

- But we can't add poses and vectors.
- They are different things.

$$ {}^{W}\mathbf{p}_d = {}^{W}\xi_C \cdot {}^{C}\mathbf{p}_d $$

transform vector from one frame to another



65

Relative Pose

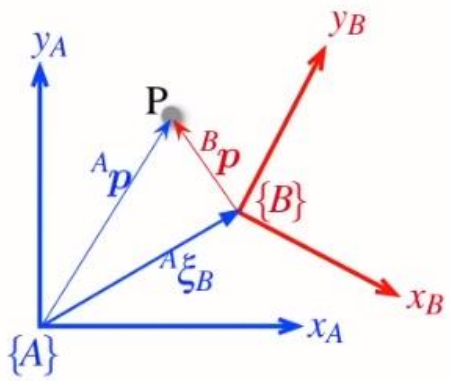
Describing relationships between the frames.

Describing a point, with a vector with respect to any frame.

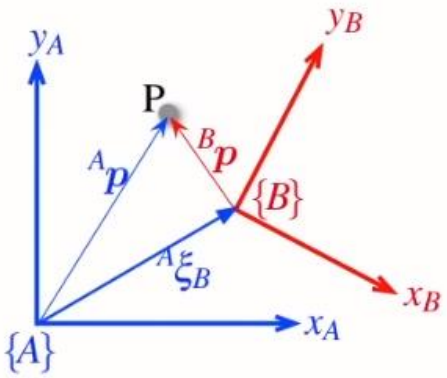


66

Relative points



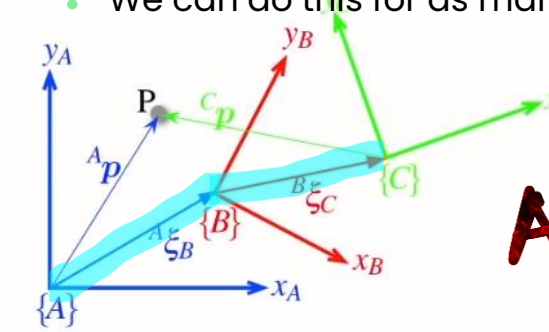
Relative points



$$ {}^A\boldsymbol{p} = {}^A\xi_B \cdot {}^B\boldsymbol{p} $$

Relative points

- Relative pose can be compounded or composed.
- We can do this for as many poses as we want.



composition

$$^A\xi_C = {}^A\xi_B \oplus {}^B\xi_C$$

$$^A p = {}^A\xi_C \cdot {}^C p$$

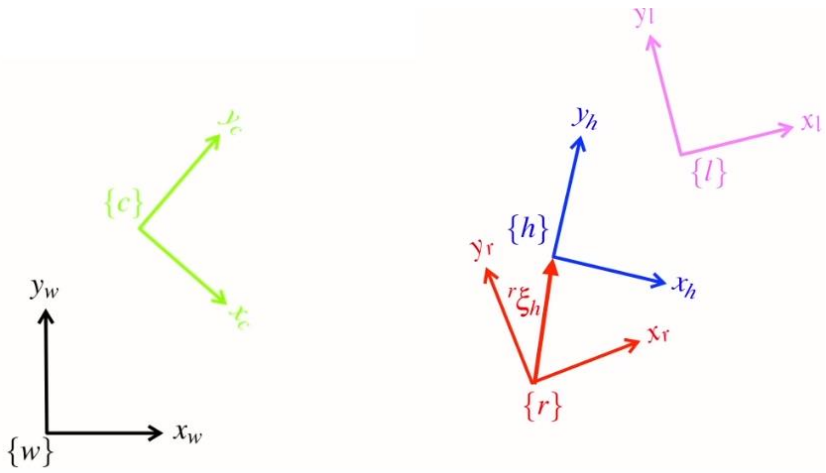
$$^A\boldsymbol{p} = ({}^A\xi_B \oplus {}^B\xi_C) \cdot {}^C\boldsymbol{p}$$

Order Matters!

Illustration

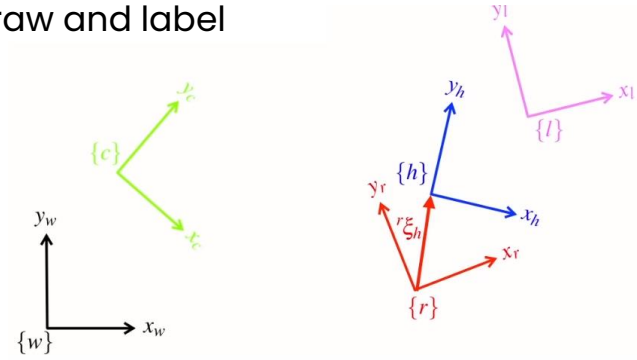
- Ordering matters!
- Tr + rot
- Rot + tr

Complex Example



Activity

- Where is robot wrt the camera?
- It means pose of robot frame wrt to camera frame.
- Draw and label



Activity

- Where is little robot wrt the camera?
- It means pose of little robot frame wrt to camera frame.
- Draw and label

